Summary of paper 01: tables and figures toward an optimal data-generating process

pmsimstats team

November 11, 2026

Problem and approach

Validating a predictive biomarker requires an interaction test. In an aggregated N-of-1 design every participant is observed on and off drug, and the power of that test must be estimated by simulation. Paper 01 asks what the data-generating process (DGP) of such a simulation should be. The DGP is a multivariate normal model in which the predictive biomarker is a correlation with the drug response (Hendrickson et al. 2020). The paper takes the open elements of the DGP one at a time, compares the candidate choices, and judges them against three requirements: a valid (positive-definite) joint distribution, an interaction that survives carryover, and flexibility to impose stress conditions. The choices are collected as the recommended DGP, a step toward an optimal one. The Hybrid design is the primary case. All simulations use the default parameter values. The biomarker is assumed throughout to be predictive only, not prognostic; the prognostic case is the subject of a companion paper. A separate section lists the differences from the DGP of Hendrickson et al. This summary reproduces every table and figure of the paper. They follow the paper's order, except that the table of matrices that are not positive definite, in the paper's Appendix B.2, appears here in Stage 1. Each is generated by the same code as in the paper.

Methods

Table 1 states which analysis targets which quantity. The paired-difference statistic E9 is the one with a closed form.

Table 1: The analyses used in this paper and the quantity each estimates.

Analysis	Drug indicator	Estimates	Used in
Paired-difference statistic E9 (closed form)	binary	on-minus-off contrast, $\beta_{bm:D}$ times the on/off correlation difference	Stages 1, 2, 4, 5; planning
The binary-indicator analysis (mixed model)	binary	the same contrast	Study 1; Differences from Hendrickson et al.
Study 2: the binary-indicator analysis	binary	the on-minus-off contrast	Stage 6
Study 2: fixed-half-life analyses	$D_{bc,it}$ at an analysis half-life of 0.5 week, the true half-life except in the cells that vary it	on-drug moderation $\beta_{bm:D}$ where the half-lives match	Stage 6
Study 2: analyses with the half-life chosen by AIC	$D_{bc,it}$ at the AIC-chosen half-life (0, 0.25, 0.5, 1 or 2 weeks)	on-drug moderation $\beta_{bm:D}$ where the chosen half-life is the true one	Stage 6
G10	$D_{bc,it}$ at the half-life chosen by AIC without the interaction (0.25, 0.5, 1 or 2 weeks)	on-drug moderation $\beta_{bm:D}$ where the chosen half-life is the true one	Stage 2

Table 2 lays out the two simulation studies. They verify the closed forms and stress-test the optional features of the recommended DGP. A further run simulates the analysis G10 for the power grid of Stage 2.

Table 2: Design of the two simulation studies in the ADEMP structure. MCSE: Monte Carlo standard error of a rejection rate, $\sqrt{p(1-p)/R}$ for R replicates.

	1. Closed-form verification	2. Stress test of the optional features
Aim	Check the closed-form moments and power of Appendix A, and the loss of power under the constant rule	Find what the optional features of the recommended DGP do to size and power
Data-generating mechanisms	The correlation encoding with the constant rule and with the decaying rule; compound symmetry and the separable AR(1) form; carryover on the drug-response mean decaying from the last on-drug value; Hybrid, crossover and OL+BDC designs; $N = 70$; grids of c_{bm} and $t_{1/2}$	The recommended DGP (Table 9), run three times: the reported run as defined, the first with the compounding carryover rule and the second with randomized allocation; base cell separable AR(1), $\rho = 0.7$, $N = 70$, $t_{1/2} = 0.5$ week; 11 null cells and alternative cells: power curves under the separable form (c_{bm} from -0.30 to 0.45, 7 cells) and under compound symmetry ($\rho_{cs} = 0.8$, c_{bm} from -0.20 to 0.25, 6 cells), and cells at $c_{bm} = 0.30$ that each vary one feature (N, half-life, decay form, dropout or measurement error). The study also ran cells with a prognostic biomarker; they are reported in the companion paper
Estimand	$\beta_{bm:D}$, and the mean of the paired-difference statistic	$\beta_{bm:D}$
Methods	Paired-difference statistics E9 and E9b, closed form and simulated; the binary-indicator mixed model fitted with `lmer`	Fourteen analyses fitted to the same simulated trials, of which ten are reported; two-sided tests at $\alpha = 0.05$

	1. Closed-form verification	2. Stress test of the optional features
Performance measures	Mean, SD and power of the statistic; agreement as (simulated − closed form)/MCSE	Null rejection rate; power; failed fits
Replicates	5,000 per cell (MCSE of power at most 0.007); 500 to 1,000 for `lmer`	1,000 per null cell (MCSE 0.0069 at a rate of 0.05), 500 per alternative cell (at most 0.022)

Results

Stage 1. Feasibility of the joint distribution

Table 3 counts correlation matrices that are not positive definite under the recommended DGP. At $c_{bm} = 0.6$, 75% of them are infeasible. The recommended DGP stops on such a matrix rather than repairing it. The choice rests on requirement 1, a valid joint distribution.

Table 3: Correlation matrices that are not positive definite under the recommended DGP (decaying biomarker-response correlation, separable AR(1)). One matrix per design path and half-life: open label plus blinded discontinuation, crossover and Hybrid designs; $t_{1/2} \in \{0, 0.1, 0.2, 0.5, 1\}$ week. The counts are the same at $\rho = 0.7$ (the recommended value) and $\rho = 0.8$ because both feasibility bounds lie between 0.3 and 0.6.

c_{bm} (number of matrices)	Recommended DGP, $\rho = 0.8$	Recommended DGP, $\rho = 0.7$
$c_{bm} = 0.25$ (40)	0 (0%)	0 (0%)
$c_{bm} = 0.3$ (40)	0 (0%)	0 (0%)
$c_{bm} = 0.6$ (40)	30 (75%)	30 (75%)

Stage 2. The decaying biomarker-response correlation

Table 4 gives closed-form power by half-life. Under the constant rule E9 power falls to the test size at every positive half-life. Under the decaying rule power stays near its no-carryover value at 0.1 to 0.2 week and declines smoothly after. The choice of the decaying rule rests on requirement 2, an interaction that survives carryover.

Table 4: Closed-form power of the paired-difference statistic E9 by carryover half-life (weeks). Hybrid design, $N = 70$, $c_{bm} = 0.25$, compound symmetry. Constant: the constant rule (constant correlation after first exposure). Decaying: the decaying rule (correlation decaying at the carryover half-life). Against Monte Carlo estimates (5,000 replicates per cell), 0 of the 12 cells in this table differ by more than 2 Monte Carlo SE, and 16 of all 566 cell-variants of study 1 (about 26 expected by chance).

Correlation rule	0	0.1	0.2	0.5	1	2
Constant	0.721	0.050	0.050	0.050	0.050	0.050
Decaying	0.721	0.721	0.712	0.631	0.473	0.278

Figure 1 shows the same result as lines, with Monte Carlo points on them.

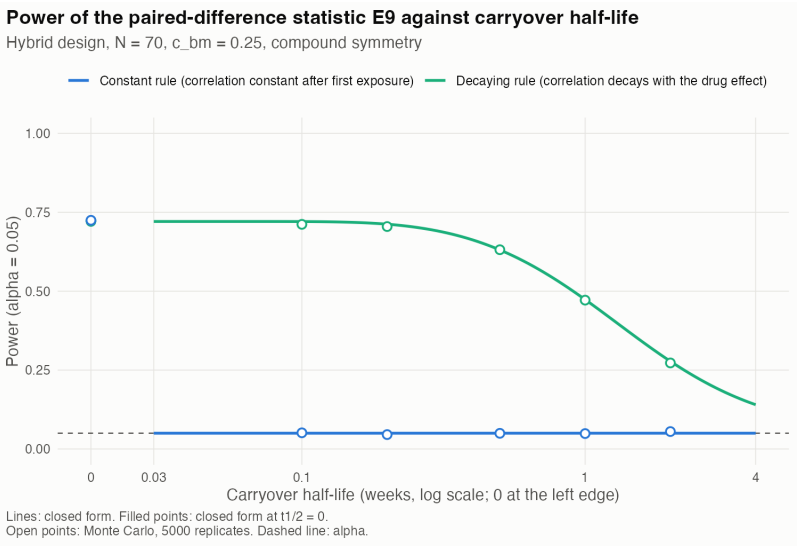


Figure 1: Closed-form power (lines) against carryover half-life, with Monte Carlo estimates (points; 5,000 replicates per cell), under the constant rule (constant correlation after first exposure) and the decaying rule (correlation decaying at the carryover half-life). Hybrid design, $N = 70$, $c_{bm} = 0.25$, compound symmetry; statistic E9.

Figure 2 plots power against the on/off correlation difference. Power depends on the correlation rule only through this difference.

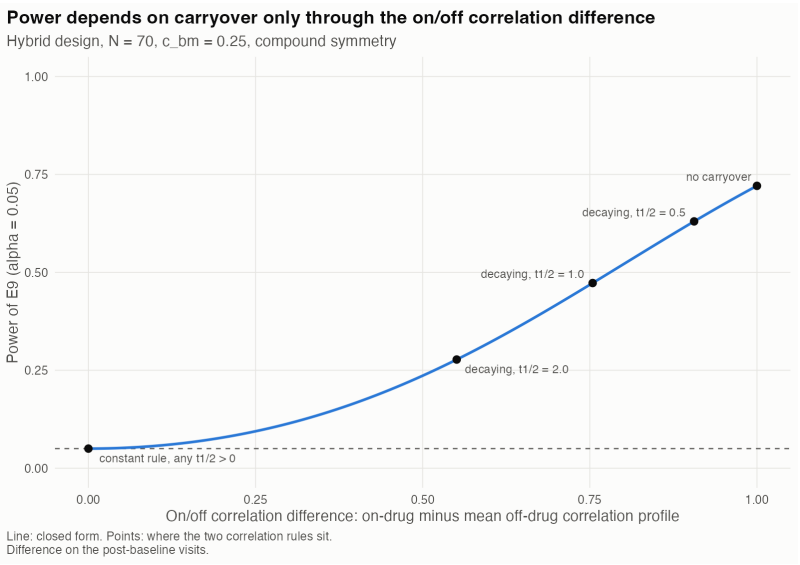


Figure 2: Closed-form power against the on/off correlation difference, with the positions of the two correlation rules. Hybrid design, $N = 70$, $c_{bm} = 0.25$, E9.

Figure 3 shows the power of the recommended analysis G10 (a random-intercept CAR(1) mixed model with an analysis half-life chosen by AIC on the model without the interaction, and CR2 standard errors) under the recommended DGP. At $N = 70$ without censoring, Hybrid power is 0.34, 0.38 and 0.34 at half-lives of 0, 0.1 and 0.2 week. The open-label design has no off-drug visit, so the interaction is not estimable under G10.

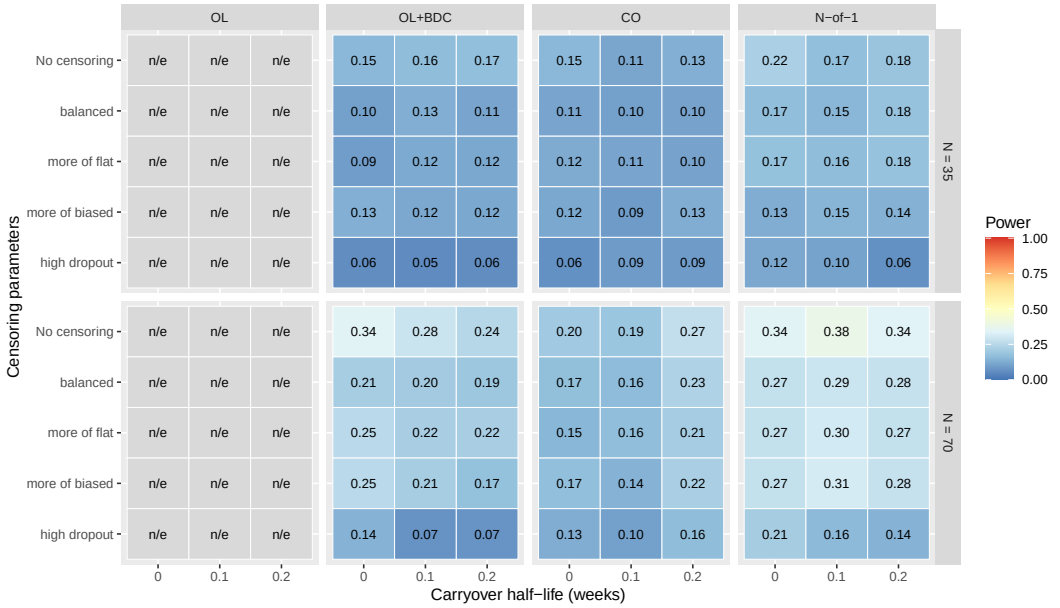


Figure 3: Power of the analysis G10 to detect the biomarker-by-treatment interaction under the recommended DGP (separable AR(1), $\rho = 0.7$, $c_1 = 0.2$, decaying biomarker-response correlation, carryover decaying from the last on-drug mean, default values). Rows within panels: censoring patterns, under which dropout rises with worsening; columns: carryover half-life; panels: trial design (N-of-1 is the Hybrid design) and total sample size. $c_{bm} = 0.25$, 250 replicates per cell (Monte Carlo SE at most 0.031). Gray cells (n/e): the open-label design, in which every visit is on drug and the interaction is not estimable under G10.

Figure 4 extends the closed-form comparison over interaction strengths and longer half-lives. Gray cells mark effect sizes with no valid joint distribution.

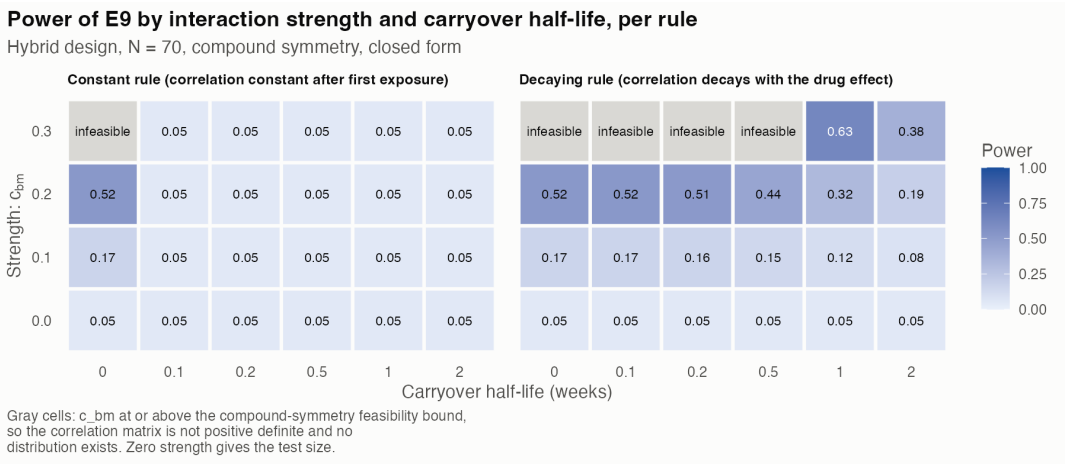


Figure 4: Closed-form power of the paired-difference statistic E9 by interaction strength (c_{bm}) and carryover half-life: the constant rule and the decaying rule. Gray cells: c_{bm} at or above the compound-symmetry feasibility bound, where the correlation matrix is not positive definite. Hybrid design, $N = 70$, compound symmetry.

Stage 3. Carryover on the mean

Figure 5 shows the drug-response mean by visit for the four Hybrid paths under both carryover rules.

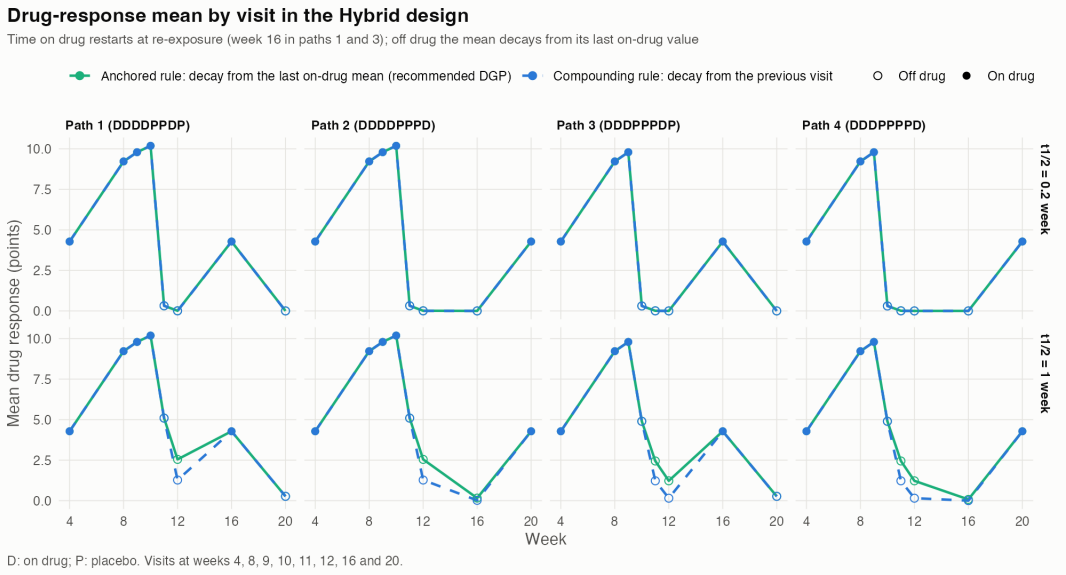


Figure 5: Drug-response mean by visit in the Hybrid design, by path, under the anchored carryover rule (decay from the last on-drug mean, the recommended DGP) and the compounding carryover rule (compounded decay), at carryover half-lives of 0.2 and 1 week. Filled points: on-drug visits; open points: off-drug visits. D: drug; P: placebo.

Stage 4. Serial (AR(1)) correlation

Table 5 shows what the serial correlation does to the feasibility bound and to power.

Table 5: Serial correlation: the feasibility bound on c_{bm} (Ceiling; decaying rule, no carryover, minimum over the Hybrid, crossover and OL+BDC designs) and closed-form E9 power (Hybrid, $N = 70$, no carryover) by ρ, under compound symmetry and under the separable AR(1) form of Stage 5. Under compound symmetry the Hybrid and crossover designs tie for the minimum. Under the separable form the minimum is attained in the Hybrid design at $\rho = 0.8$ and in the OL+BDC design at $\rho \leq 0.7$, and the crossover design is never binding. At $t_{1/2} = 0.2$ week every power value is lower by 0.004 to 0.009.

	Compound symmetry		Separable AR(1)	
ρ_{cs} or ρ	Ceiling	Power at $c_{bm} = 0.25$	Ceiling	Power at $c_{bm} = 0.25$
0.5	0.458	0.492	0.448	0.261
0.6	0.410	0.553	0.462	0.270
0.7	0.346	0.628	0.474	0.290
0.8	0.256	0.721	0.440	0.335

Table 6 extends the separable form to ρ from 0.4 to 0.95. The bound is largest at 0.7 and falls steeply above it.

Table 6: Serial correlation under the separable AR(1) form of the recommended DGP ($c_1 = 0.2$, decaying rule, carryover decaying from the last on-drug mean): the feasibility bound on c_{bm}, minimum over the Hybrid, crossover and OL+BDC designs, by carryover half-life, the design that attains the minimum without carryover, and the closed-form total N for 80 percent power of E9 in the Hybrid design at the 0.2-week half-life. Not feasible: c_{bm} lies above the Hybrid bound.

	Feasibility bound				Hybrid N, E9	
ρ	$t_{1/2} = 0$	0.2 week	1 week	Binding design	$c_{bm} = 0.25$	$c_{bm} = 0.35$
0.40	0.437	0.438	0.445	OL+BDC	318	160
0.50	0.448	0.450	0.465	OL+BDC	314	158
0.60	0.462	0.465	0.494	OL+BDC	302	152
0.65	0.469	0.473	0.512	OL+BDC	292	148
0.70	0.474	0.479	0.533	OL+BDC	278	140
0.75	0.465	0.471	0.550	Hybrid	258	130
0.80	0.440	0.446	0.544	Hybrid	234	118
0.85	0.400	0.407	0.519	Hybrid	200	100
0.90	0.340	0.346	0.457	Hybrid	154	not feasible
0.95	0.247	0.251	0.343	Hybrid	92	not feasible

Stage 5. The separable cross-component structure

Table 7 compares three correlation structures.

Table 7: Largest feasible biomarker-response correlation c_{bm} under the decaying rule, without carryover, minimum over the Hybrid, crossover and OL+BDC designs. Under compound symmetry the Hybrid and crossover designs tie for the minimum. Under the two AR(1) forms the minimum is attained in the Hybrid design at $\rho = 0.8$ and in the OL+BDC design at $\rho = 0.7$ (separable form: Hybrid 0.480), and the crossover design is never binding.

ρ_{cs} or ρ	Compound symmetry	AR(1), two-constant cross-component	Separable AR(1)
0.8	0.256	0.291	0.440
0.7	0.346	0.452	0.474

Stage 6. Optional features

Table 8 shows what each optional feature does to size and power, in the third run of study 2, the recommended DGP with fixed allocation.

Table 8: The optional features of the recommended DGP: null rejection rate ($c_{bm} = 0$; 1,000 replicates, MCSE 0.007) and power ($c_{bm} = 0.30$; 500 replicates, MCSE at most 0.022) for the binary-indicator analysis with the working-covariance standard error (Binary), and the random-intercept CAR(1) model and the expectancy-term model (Expect.), both with the cluster-robust standard error (CR2). Hybrid design, third run of study 2 (the recommended DGP: carryover decaying from the last on-drug mean, fixed allocation). Each row changes one feature of the base cell.

	Null rejection			Power		
Cell	Binary	CAR(1)	Expect.	Binary	CAR(1)	Expect.
Base cell ($N = 70$, $t_{1/2} = 0.5$, exponential, no dropout)	0.077	0.051	0.067	0.474	0.486	0.404
Measurement error (SD 3 and 6)	0.069	0.050	0.037	0.422	0.394	0.314
Dropout 20 percent, completely at random	0.078	0.053	0.055	0.466	0.464	0.392
Dropout 20 percent, dependent on last score	0.069	0.060	0.047	0.424	0.426	0.360
Weibull decay, shape 0.5	0.072	0.046	0.044	0.378	0.400	0.300
Weibull decay, shape 2	0.066	0.039	0.046	0.526	0.540	0.496
$t_{1/2} = 0$	0.071	0.040	0.041	0.536	0.566	0.498
$t_{1/2} = 1$ week	0.066	0.047	0.061	0.418	0.416	0.348
$N = 35$	0.077	0.053	0.048	0.270	0.246	0.206
$N = 150$	0.060	0.048	0.045	0.768	0.790	0.688

Stage 7. The recommended DGP

Table 9 collects the choices. Each row is marked as a choice made in a stage, a default value, an option, or an assumption, and names the section that justifies it.

Table 9: The recommended data-generating process. The Default column gives the value of each element in the recommended DGP; where the stress test of Stage 6 used another value it is stated. The stress test's reported run uses fixed allocation; its second run assigned participants to paths at random, a driver option outside the recommended DGP (Appendix B.5).

Element	Specification	Default	Status	Justified in
Components	$Y_{it} = \mathrm{BL}_i - (TV_{it} + PB_{it} + BR_{it})$, modified Gompertz means	default values (Methods)	default	Methods
Feasibility	stop if the correlation matrix is not positive definite	always	choice	Stage 1
Decaying biomarker-response correlation	c_{bm} on drug; $c_{bm} 2^{-t_{sd}/t_{1/2}}$ off drug after exposure; 0 before exposure	c_{bm} below the feasibility bound of every design in the study	choice	Stage 2
Carryover on the mean (anchored)	decay from the last on-drug value, $\mu_{BR,\mathrm{last}}\, 2^{-t_{sd}/t_{1/2}}$	$t_{1/2} = 0.1$ to 0.2 week (pharmacological, assumed); 0.5 in the stress test	choice	Stage 3
Re-exposure	time on drug restarts at zero	on, for titrated drugs	choice	Stage 3
Serial (AR(1)) correlation	$\rho^{\lvert w_t - w_s \rvert}$ in calendar weeks, one ρ for all three components	$\rho = 0.7$ (convention; report and vary)	choice	Stage 4
Separable cross-component structure	$c_1 \rho^{\lvert w_t - w_s \rvert}$ at every lag, applied to the correlation, since the PB SD scales with the expectancy	$c_1 = 0.2$ (default value)	choice	Stage 5

Element	Specification	Default	Status	Justified in
Path allocation	fixed numbers	18, 18, 17, 17 at $N = 70$ (Hybrid)	default	Methods
Prognostic association	none: the biomarker is correlated with BR only, not with TV, PB or baseline severity	zero (the prognostic case is treated in the companion paper)	assumption	Aims
Measurement error	independent, SD by phase	off; 3 open label, 6 blinded in the stress test	option (driver only)	Stage 6
Decay form	exponential or Weibull with the same half-life	exponential	option	Stage 6
Dropout	monotone, completely at random or at random given the last score	off; 20% in the stress test	option	Stage 6

Using the recommended DGP to plan a trial

Table 10 gives the closed-form total sample size for 80 percent power of the paired-difference statistic E9. Sample sizes for the recommended mixed-model analysis G10 are in the companion paper. Table 11 repeats the calculation at $\rho = 0.5$ and 0.9. At 0.9 the Hybrid design needs about half the sample.

Table 10: Total N for 80 percent power of the paired-difference statistic E9 at $\alpha = 0.05$, in closed form, under the recommended DGP (separable AR(1), $\rho = 0.7$, $c_1 = 0.2$, decaying biomarker-response correlation, carryover decaying from the last on-drug mean, default values), by design, interaction strength and carryover half-life, with the implied slope in symptom points per SD of biomarker ($\sigma_{BR} = 8$). Without carryover the values are at most 3 percent below those at 0.2 week. Participants are allocated to paths in fixed proportions (18, 18, 17, 17 in the Hybrid design at $N = 70$). Sample sizes for the mixed-model analysis G10 are in the companion paper.

			Carryover half-life	
Design	c_{bm}	Slope (points per SD)	0.2 week	1 week
Hybrid	0.25	2.0	278	480
	0.35	2.8	140	244
	0.45	3.6	84	146
OL+BDC	0.25	2.0	404	> 800
	0.35	2.8	206	452
	0.45	3.6	124	272
Crossover	0.25	2.0	424	454
	0.35	2.8	216	230
	0.45	3.6	130	138

Table 11: Total N for 80 percent power of E9 in closed form by serial correlation, as Table 10 (which gives $\rho = 0.7$) but at $\rho = 0.5$ and 0.9, at carryover half-lives of 0.2 and 1 week. At $\rho = 0.9$ the grid of c_{bm} lies inside the lower bound. Not feasible: c_{bm} lies above the bound of that design.

			Carryover half-life	
ρ	Design	c_{bm}	0.2 week	1 week
0.5	Hybrid	0.25	314	542
		0.35	158	276
		0.45	96	166
	OL+BDC	0.25	412	$>$ 800
		0.35	208	460
		0.45	126	278
	Crossover	0.25	348	372
		0.35	176	188
		0.45	106	114
0.9	Hybrid	0.20	242	420
		0.25	154	268
		0.30	106	186
	OL+BDC	0.20	462	$>$ 800
		0.25	296	650
		0.30	204	450
	Crossover	0.20	628	672
		0.25	402	430
		0.30	278	298

Differences from the DGP of Hendrickson et al.

Table 12 lists every difference between the recommended DGP and the DGP of Hendrickson et al. (their code at commit 58b32a9).

Table 12: Differences between the DGP of Hendrickson et al. (code at commit 58b32a9) and the recommended DGP of this paper.

Element	Hendrickson et al.	This paper	Justified in
Infeasible matrices	replaced by a nearby valid matrix (Higham's nearest positive semidefinite matrix, tolerance 10^{-3}), without a warning	refused: the cell stops and is reported	Stage 1; Appendix B.2
Correlation off drug	constant: c_{bm} wherever the drug-response mean is nonzero, so the full value at every visit after first exposure	decaying: $c_{bm} 2^{-t_{sd}/t_{1/2}}$ after exposure, 0 before	Stage 2; Appendix B.1
Mean off drug	compounding: each off-drug mean decayed from the previous visit's	anchored: decayed from the last on-drug mean	Stage 3; Appendix B.3
Serial correlation	compound symmetry, $\rho_{cs} = 0.8$	AR(1) in calendar weeks, $\rho = 0.7$	Stage 4
Cross-component correlation	$c_1 = 0.2$ at the same visit, $c_\times = 0.1$ at every other lag	separable, $c_1 \rho^{\lvert w_t - w_s \rvert}$ at every lag, $c_1 = 0.2$	Stage 5
Dropout weight	weight rises with improvement (100 plus the improvement)	weight rises with worsening (100 minus the improvement, floored at zero)	Appendix B.5
Optional features	none	measurement error, Weibull decay, dropout	Stage 6

Table 13 reruns their code as published and with only the correlation lines replaced by the decaying rule. As published, power falls from 0.668 to 0.126 at 0.1 week; a rerun of that cell with 4,000 new replicates gave power 0.090. With the decaying rule the interaction is kept.

Table 13: The code of Hendrickson et al. at commit 58b32a9, as published (constant) and with the decaying biomarker-response correlation (decaying): Hybrid design, $N = 70$, $c_{bm} = 0.25$, binary-indicator analysis, 500 replicates per cell, same seeds. Monte Carlo SE of power at most 0.022. A rerun of the published version at 0.1 week with 4,000 new replicates gave power 0.090.

$t_{1/2}$ (weeks)	Mean estimate, constant	Mean estimate, decaying	Power, constant	Power, decaying
0	-0.131	-0.131	0.668	0.668
0.1	-0.028	-0.129	0.126	0.638
0.2	-0.029	-0.131	0.092	0.656
1	-0.033	-0.107	0.094	0.462

Figure 6 repeats the comparison for all four designs, both sample sizes and their five censoring patterns.

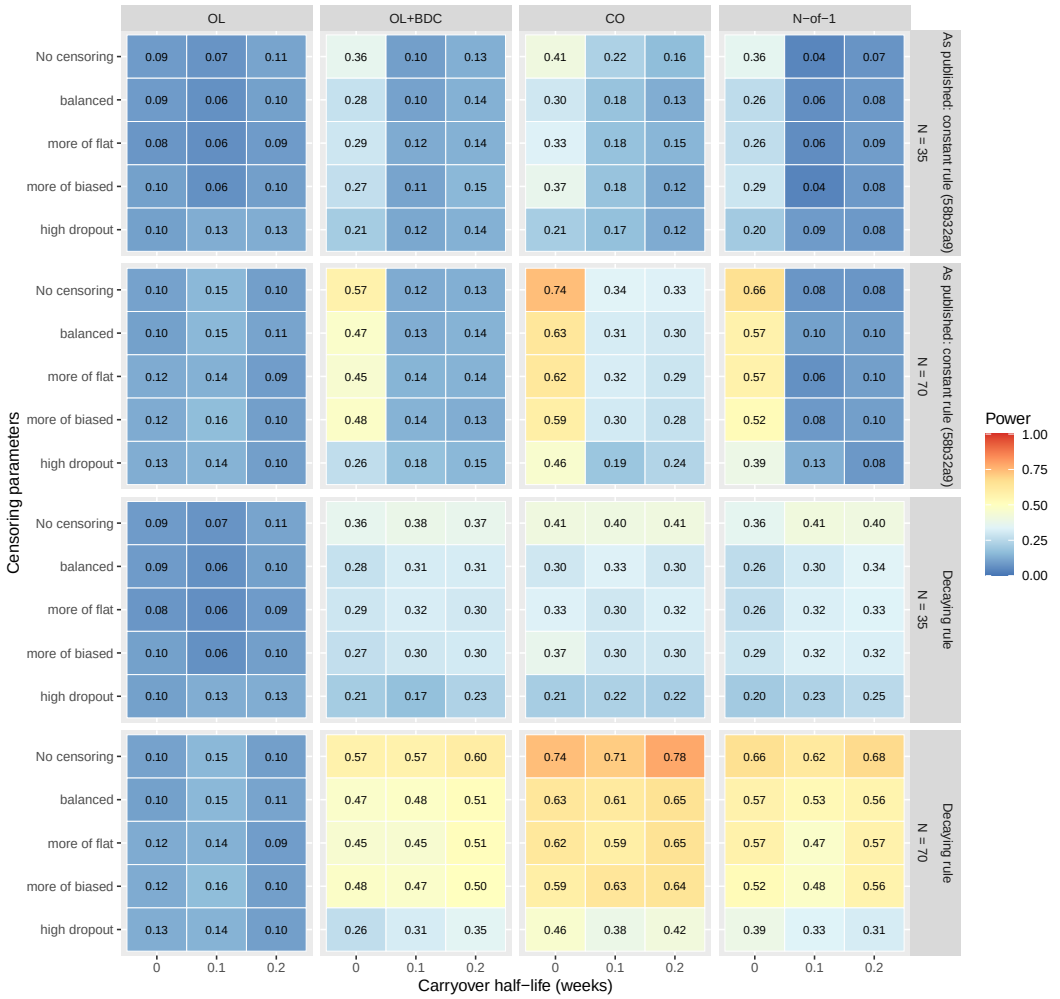


Figure 6: Power to detect the biomarker-by-treatment interaction from the code of Hendrickson et al. at commit 58b32a9, in the layout of their Figure 4B, as published (upper two panel rows) and with the decaying biomarker-response correlation (lower two panel rows). Rows within panels: their censoring patterns; columns: carryover half-life; panels: trial design (N-of-1 is the Hybrid design) and total sample size. $c_{bm} = 0.25$, binary-indicator analysis, 250 replicates per cell (Monte Carlo SE at most 0.032), the same seeds for both versions. No matrix was repaired in either version.

Appendix A. Closed-form derivations

A.5 The separable structure

Each step between visits contributes the following term to the schedule term of the feasibility bound.

Step	Term
on to on	$(1-\rho_t)/(1+\rho_t)$
on to off	$\rho_t^2/(1-\rho_t^2)$
off to on	$1/(1-\rho_t^2)$
off to off	0

Conclusions

Three choices matter most. First, the biomarker-response correlation must decay after discontinuation. If it stays constant after first exposure, the interaction is absent among the post-baseline visits under any carryover, and power falls nearly to the test size. With a decaying biomarker-response correlation, power returns to its no-carryover value at pharmacological half-lives and declines smoothly at half-lives of 0.5 week and above. Second, the serial correlation structure sets the largest interaction the DGP can hold. A separable AR(1) structure at a serial correlation of 0.7 raises the bound from 0.256 under compound symmetry to 0.474. The value 0.7 is a convention, however. At 0.9 the bound falls to 0.340, and the Hybrid design needs 154 participants at $c_{bm} = 0.25$ instead of 278. Third, a matrix outside the bound should be refused rather than repaired.

We propose the recommended DGP as a reference for simulation studies of these designs when the biomarker is predictive only. Every parameter is labeled as a default value, a choice or uncalibrated. Calibrating the serial correlation, the drug-response SD and the symptom half-life is the next step toward an optimal DGP.